

The intended Steklov exponential-localization conjecture is answered

Literature status and statement correction for arXiv:0801.4145

Autonomous mathematical discovery run `fa_banach_001`

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Status: `literature_already_answered`, with an important correction to the source formulation.

Source question

Zagrebnov’s Conjecture 2.7 on PDF page 6 of [1] attributes to Hislop and Lutzer the claim that harmonic extensions of Dirichlet-to-Neumann eigenfunctions have exponential interior decay. The display in that paper reads

$$O(\exp[-k \operatorname{dist}(\mathcal{C}, \partial\Omega)]) .$$

Explicit later answer

Galkowski and Toth formulate the spectral problem semiclassically as

$$(h\mathcal{D} - 1)\varphi_h = 0,$$

so the spectral frequency is h^{-1} . In their introduction they explicitly write that their main result proves the Hislop–Lutzer conjecture in arbitrary dimension [2]. Their Theorem 1, on PDF pages 2–3, proves for compact real-analytic manifolds with real-analytic boundary a sharp near-boundary estimate of the form

$$|\partial_x^\alpha u_h(x)| \leq C_{\alpha,\delta} h^{-N_\alpha} \exp\left(-\frac{d(x)}{h}\right)$$

in the stated tubular region, together with a global interior estimate. This is the intended exponential localization, with rate governed by the Dirichlet-to-Neumann eigenvalue.

Statement correction and scope

The standalone wording in [1] omits the real-analytic hypothesis present in the Hislop–Lutzer conjecture and uses k immediately after k has denoted ordinal eigenvalue position. Those are material differences:

- Galkowski–Toth stress that analyticity is needed for exponential accuracy; smooth calculus gives only $O(h^\infty)$ control.
- The decay rate is spectral frequency $h^{-1} = \lambda$, not ordinal index. On a three-dimensional ball, degree- l spherical harmonics have eigenvalue l/R with multiplicity $2l + 1$, so ordinal index is of order l^2 , while the harmonic lifting decays like $(r/R)^l$.

Indeed, the ball example immediately following Conjecture 2.7 in [1] obtains $O(e^{-l(R-r)/R})$, which is consistent with spectral frequency and not with the ordinal index after multiplicities are counted. Accordingly, the analytically corrected intended conjecture is answered by [2]; the literal smooth/index sentence in the 2008 source should not be quoted as the theorem they proved.

Other extracted signal

The trace-norm product-approximation conjecture later in [1] is followed in the same section by the statement that the displayed argument proves it “at least for the ball.” That ball case is a same-paper ask-and-answer passage, not a separate literature-answer target. Any genuinely general-domain version would require a new precise formulation of the approximating family and uniform trace estimates.

Search record

The identification was checked on 17 August 2026 against the official arXiv records, the original source text, the introduction and Theorem 1 of arXiv:1611.05363, and a modern Steklov survey that likewise states that Galkowski–Toth proved the Hislop–Lutzer conjecture in all dimensions.

References

- [1] V. A. Zagrebnov, *From Laplacian Transport to Dirichlet-to-Neumann (Gibbs) Semigroups*, arXiv:0801.4145 (2008).
- [2] J. Galkowski and J. A. Toth, *Pointwise Bounds for Steklov Eigenfunctions*, J. Geom. Anal. 29 (2019), 142–193; arXiv:1611.05363.